

Hossein Khayami

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Education

University of Maryland, College Park, MD

Expected May 2027

PhD Student in Electrical Engineering - Communication and Signal Processing

– GPA (up to now): 3.71/4.0

Sharif University of Technology, Tehran

Sep 2015

Master of Science in Electrical Engineering - Communication Systems

– Average: 17.59/20 (4.0/4.0)

University of Tehran, Tehran

Feb 2013

Bachelor of Science in Electrical Engineering - Telecommunications

– Average: 16.64/20 (3.5/4.0)

Skills

Machine Learning TensorFlow, TFLite, PyTorch, Scikit-learn, Keras, CNNs, RNNs, LSTMs, Model Compression, Quantization, Optimization

Embedded Systems RTOS, Bare-metal firmware, IoT Devices

Data Analysis SQL, Pandas, Numpy
Programming Python, MATLAB, C/C++
Signal Processing Sensor Fusion, Real-time DSP

Hardware Circuit and PCB Design (Altium)

Publications

H. Khayami, “Personalizing Activity Tracking: Enabling Older Adults to Teach and Train Their Wearables,” In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland.

A. Momtaz, E. Qoreishi, S. Amini, **H. Khayami**, K. Amini, and S. Haddadian, “Chaos-based Pseudo Random Number Generators via quasi-synchronized Chua’s circuits: a symmetric encryption perspective,” *Analog Integr Circ Sig Process*, vol. 126, no. 26, p. 19, Dec. 2025.

Y. Wang, **H. Khayami**, B. Lee, A. Lazar, H. Kacorri and E. Choe, “Enabling Older Adults to Provide High-quality Activity Labels: Unpacking Accuracy, Precision, and Granularity in Activity Labeling,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 4, Article 222 (December 2025), 24 pages.

H. Khayami, L. Wang, Y. Kim, B. Lee, D. Conroy, A. Lazar, E. Choe and H. Kacorri, “From Verbal Reports to Personalized Activity Trackers: Understanding the Challenges of Ground Truth Data Collection with Older Adults in the Wild,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 9, 2, Article 38 (June 2025), 33 pages.

V. Ramani, **H. Khayami**, Y. Bai, N. Garg and N. Roy, “IMUOptimize: A Data-Driven Approach to Optimal IMU Placement for Human Pose Estimation with Transformer Architecture,” *arXiv preprint arXiv:2402.08923* (2024).

H. Khayami, T. Eghlidos and M.R. Aref, “A Joint Encryption-Encoding Scheme Using QC-LDPC Codes Based on Finite Geometry,” *Scientia Iranica* (2024) 31(17), pp. 1504-1516

M. Shirvanimoghaddam, **H. Khayami**, Y. Li and B. Vucetic, “Dynamic HARQ with Guaranteed Delay,” *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, Seoul, Korea, May 2020.

H. Khayami, M. Ghassemi, K. Ardekani, B. Maham and W. Saad, “Cognitive Radio Ad Hoc Networks for Smart Grid Communications: A Disaster Management Approach,” *2013 IEEE/CIC International Conference on Communications in China (ICCC)*, pp.716-721, Aug. 2013.

H. Morsali, S. M. Shekarabi, K. Ardekani, **H. Khayami**, A. Fereidunian, M. Ghassemian and H. Lesani, “Smart Plugs for Building Energy Management Systems,” *2nd Iranian Conference on Smart Grids (ICSG 2012)*, May 24-25, Tehran.

Research Experience

Intelligent Assistive Machines (IAM) Lab, University of Maryland, College Park, MD

Jun 2023 – present

Research Assistant

- Developing personalized machine learning models with teachable interactions to make wearable activity tracking more accessible and adaptive for older adults.
- Investigated novel activity data collection methods combining self-reports and sensor-based monitoring, advancing understanding of real-world physical activity patterns in older adult populations.

Professional Experiences

Vicinia, California <i>IoT Network Engineer</i>	May 2022 – Aug 2022
– Designed and developed hardware, firmware, and communication protocols for a wireless IoT data gathering node in a cloud-based indoor Real-Time Locating System (RTLS). – Enabled on-device and on-server positioning for campus navigation and asset tracking. – Reduced network coverage costs by 50% through an optimized two-layer edge architecture, lowering hardware and infrastructure requirements.	
MTN, Tehran <i>Data Analyst, Network Performance Engineer</i>	Mar 2020 – Jul 2021
– Automated generation of routine KQI/KPI reports and anomaly detection procedures using Python and SQL, saving several hours of manual work daily. – Analyzed core network quality and performance indicators on demand to support data-driven operational decisions.	
Arshon Technology, Ontario, Canada (remote) <i>Senior Hardware Engineer</i>	Dec 2020 – Jul 2021
– Engineered hardware and firmware for an industrial IoT gateway, enabling customers to remotely monitor and control building equipment and improving operational efficiency.	
Sarveen Technologies Inc., Tehran <i>Head of Embedded Systems Team</i>	Sep 2016 – Feb 2020
– Headed the embedded systems team delivering hardware, firmware, and wireless protocols across a full suite of AI-enabled livestock health monitoring devices — electronic milk meter, ultra-low-power wearable, IoT gateway, walk-over weigh scale, and exhale analyzer. – Owned procurement, manufacturing, and production for all hardware products; led development planning and time-to-market estimation in collaboration with product managers and the sales team. – Collaborated with ML and software teams to deploy efficient on-device algorithms on firmware, and with the industrial design team to align mechanical and electronic product requirements. – Developed signal processing algorithms and C/C++ firmware for IMU-based wearable activity recognition and electronic milk metering, improving measurement reliability.	

Teaching Experiences

University of Maryland <i>Teaching Assistant</i>	2021 – 2026
– Signal and System Theory and Applications, Cryptography, Digital Circuits and Systems Laboratory, Embedded Systems, Convex Optimization, Introduction to Networks and Protocols	
Sharif University of Technology <i>Teaching Assistant</i>	2015
– Data Communication Networks	
University of Tehran <i>Teaching Assistant and Instructor</i>	2011 – 2013
– Wireless Multimedia Communications lab, Signal and Systems, Microprocessors	
Exceptional Talent High Schools <i>Teacher</i>	2008 – 2015
– Principles of Computer Programming, Robotics and Embedded Programming, Life and Social Skills	

Professional Services

Peer-Reviews	
– ACM IMWUT (UbiComp)	2025
– IEEE MCSoc (Multicore and Many-core Systems-on-Chip)	2024
– IEEE Wireless Africa Conference, <i>The IEEE Vehicular Technology Society</i>	2019
– Physical Communication Journal, <i>Elsevier</i>	2019
Referee and Technical Committee	
– Internet of Things Challenge, <i>Iranian University of Science and Technology</i>	2017
– Kharazmi Innovation Festival of Youths for Electronics projects	2017
– RoboCup IranOpen International Competitions in Junior Leagues	2007 – 2015
Student Volunteer	
– Human-Computer Interaction Lab (HCIL) Symposium	2023 – 2024
– 20th Iranian Conference on Electrical Engineering	2012